

ASCEND Life RPG - Developer & Maintenance Handbook

Reference manual for maintaining, updating, and extending the project over the coming years.

1. Core Architecture

index.html = Application shell and script loading order.
css/main.css = All styling and UI appearance.
js/data.js = Stats, ranks, achievements, defaults, constants.
js/engine.js = Core game logic and save handling.
js/app.js = Main controller and navigation.
js/pages/* = Individual pages and UI rendering.

2. Save System

Save data is stored in browser localStorage under the key 'ascend_v1'. Never change this key unless you intentionally want a new save format. Before major updates export a backup.

3. Safe Updates

Safe: CSS changes, new quests, new achievements, UI improvements, balancing XP, adding new skills.

Dangerous: deleting fields, changing save structure, renaming save keys, clearing localStorage.

4. Deployment Workflow

Standard update process:

```
git add .  
git commit -m "describe change"  
git push
```

5. PWA Cache Updates

When users do not receive updates, increment the cache version in sw.js: ascend-v1 -> ascend-v2 -> ascend-v3. Then commit and push.

6. Backup Strategy

Use Export Save before major updates. Keep periodic JSON backups. Keep a ZIP backup of the full project folder.

7. Feature Expansion Guide

New page: create js/pages/newpage.js, register in app.js, add navigation button.

New achievement: add definition in data.js.

New stat: add stat definition in data.js and render logic where required.

New quest type: add quest type in data.js and handling in engine.js.

8. Disaster Recovery

If GitHub breaks: clone/download repo and redeploy.

If phone data is lost: Import Save JSON.

If a bad update ships: restore previous Git commit and redeploy.

9. Long-Term Rules

Keep save compatibility a priority. Add features without deleting old fields. Test changes in browser before pushing. Create backups before major rewrites.